Probability and Statistics

Point Estimation

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1 Some General Concepts of Point Estimation

2 Methods of Point Estimation

Some General Concepts of Point Estimation

- Point estimate (点估计值)
  - A point estimate of a parameter θ is a single number that can be regarded as an estimated value of the considered parameter θ
  - A point estimate is obtained by selecting a suitable statistic and computing its value from the given sample data
- Point estimator (点估计量)
  - The selected statistic is called the point estimator of θ

- Point estimate 是具体的估计结果，point estimator 是用来计算这个结果的工具（统计量）
- $\hat{\theta}$ can be used to denote both point estimate and point estimator

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Some General Concepts of Point Estimation

- If random variable X follows a normal probability distribution, and we collected 20 random sample shown below

24.46	25.61	26.25	26.42	26.66	27.15	27.31	27.54	27.74	27.94
27.98	28.04	28.28	28.49	28.50	28.87	29.11	29.13	29.50	30.88

- Following shows several examples of point estimators and point estimates

a. Estimator $= \bar{X}$, estimate $= \bar{x} = \sum x_i/n = 555.86/20 = 27.793$

b. Estimator $= \tilde{X}$, estimate $= \tilde{x} = (27.94 + 27.98)/2 = 27.960$

c. Estimator $= [\min(X_i) + \max(X_i)]/2 =$ the average of the two extreme lifetimes, estimate $= [\min(x_i) + \max(x_i)]/2 = (24.46 + 30.88)/2 = 27.670$

d. Estimator $= \bar{X}_{\text{tr}(10)}$, the 10% trimmed mean (discard the smallest and largest 10% of the sample and then average),

$$\begin{aligned}\text{estimate} &= \bar{x}_{\text{tr}(10)} \\ &= \frac{555.86 - 24.46 - 25.61 - 29.50 - 30.88}{16} \\ &= 27.838\end{aligned}$$

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Some General Concepts of Point Estimation

- Unbiased estimator (无偏估计量)

A point estimator $\hat{\theta}$ is said to be an **unbiased estimator** of θ if $E(\hat{\theta}) = \theta$ for every possible value of θ. If $\hat{\theta}$ is not unbiased, the difference $E(\hat{\theta}) - \theta$ is called the **bias** of $\hat{\theta}$.

Some General Concepts of Point Estimation

- Suppose we have a radom sample of size 3, $X_1 \sim X_3$, from a normal distribution with mean μ
- Determine if the following estimators of μ are biased or not
- $1/2 * X_1 + 1/4 * X_2 + 1/4 * X_2$
- $1/2 * X_1 + 1/2 * X_2 + 1/4 * X_2$

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- Determine if the following estimators of μ are biased or not
- $1/2 * X_1 + 1/4 * X_2 + 1/4 * X_2$ is an unbiased estimator
- $1/2 * X_1 + 1/2 * X_2 + 1/4 * X_2$ is a biased estimator

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Some General Concepts of Point Estimation

- Random variable X is a uniform random variable whose range is from 0 to θ and we have a random sample of X: $X_1 \sim X_n$. Now we have two different estimators of θ: $\widehat{\theta_1}$ and $\widehat{\theta_2}$, where $\widehat{\theta_1} = max(X_1 \sim X_n)$, $\widehat{\theta_2} = \frac{n+1}{n} max(X_1 \sim X_n)$
- Which one is unbiased?

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- Which one is unbiased?

$$E(\hat{\theta}_1) = \frac{n}{n+1} \cdot \theta < \theta \qquad \left(\text{since } \frac{n}{n+1} < 1\right)$$

$$E(\hat{\theta}_2) = E\left[\frac{n+1}{n} \max(X_1, \ldots, X_n)\right] = \frac{n+1}{n} \cdot E[\max(X_1, \ldots, X_n)]$$

$$= \frac{n+1}{n} \cdot \frac{n}{n+1} \theta = \theta$$

- So, $\widehat{\theta_2}$ is unbiased

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Some General Concepts of Point Estimation

- Sample variance (S^2) and standard deviation (S)

$$S^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n - 1}$$

$$S = \sqrt{S^2}$$

- Population variance (σ^2) and standard deviation (σ)

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

$$\sigma = \sqrt{\sigma^2}$$

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Some General Concepts of Point Estimation

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Then the estimator

$$\hat{\sigma}^2 = S^2 = \frac{\sum(X_i - \bar{X})^2}{n-1}$$

is an unbiased estimator of σ^2.



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Proof For any rv Y, $V(Y) = E(Y^2) - [E(Y)]^2$, so $E(Y^2) = V(Y) + [E(Y)]^2$. Applying this to

$$S^2 = \frac{1}{n-1}\left[\sum X_i^2 - \frac{(\sum X_i)^2}{n}\right]$$

gives

$$\begin{aligned}
E(S^2) &= \frac{1}{n-1}\left\{\sum E(X_i^2) - \frac{1}{n}E[(\sum X_i)^2]\right\} \\
&= \frac{1}{n-1}\left\{\sum(\sigma^2 + \mu^2) - \frac{1}{n}\{V(\sum X_i) + [E(\sum X_i)]^2\}\right\} \\
&= \frac{1}{n-1}\left\{n\sigma^2 + n\mu^2 - \frac{1}{n}n\sigma^2 - \frac{1}{n}(n\mu)^2\right\} \\
&= \frac{1}{n-1}\{n\sigma^2 - \sigma^2\} = \sigma^2 \qquad \text{(as desired)}
\end{aligned}$$

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Some General Concepts of Point Estimation

If $X_1, X_2, \ldots, X_n$ is a random sample from a distribution with mean μ, then $\overline{X}$ is an unbiased estimator of μ. If in addition the distribution is continuous and symmetric, then $\widetilde{X}$ and any trimmed mean are also unbiased estimators of μ.

1 Some General Concepts of Point Estimation

2 Methods of Point Estimation

Methods of Point Estimation

- The method of moments（矩估计法）
- *k*th population moment and sample moment

Let $X_1, \ldots, X_n$ be a random sample from a pmf or pdf $f(x)$. For $k = 1, 2, 3, \ldots$, the ***k*th population moment,** or ***k*th moment of the distribution $f(x)$,** is $E(X^k)$. The ***k*th sample moment** is $(1/n)\sum_{i=1}^{n} X_i^k$.

Methods of Point Estimation

- The method of moments（矩估计法）

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with pmf or pdf $f(x; \theta_1, \ldots, \theta_m)$, where $\theta_1, \ldots, \theta_m$ are parameters whose values are unknown. Then the **moment estimators** $\hat{\theta}_1, \ldots, \hat{\theta}_m$ are obtained by equating the first m sample moments to the corresponding first m population moments and solving for $\theta_1, \ldots, \theta_m$.

Let $X_1, X_2, \ldots, X_n$ represent a random sample of service times of n customers at a certain facility, where the underlying distribution is assumed exponential with parameter λ. Since there is only one parameter to be estimated, the estimator is obtained by equating $E(X)$ to $\bar{X}$. Since $E(X) = 1/\lambda$ for an exponential distribution, this gives $1/\lambda = \bar{X}$ or $\lambda = 1/\bar{X}$. The moment estimator of λ is then $\hat{\lambda} = 1/\bar{X}$.

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Methods of Point Estimation

Let $X_1, \ldots, X_n$ be a random sample from a gamma distribution with parameters α and β. From Section 4.4, $E(X) = \alpha\beta$ and $E(X^2) = \beta^2\Gamma(\alpha + 2)/\Gamma(\alpha) = \beta^2(\alpha + 1)\alpha$. The moment estimators of α and β are obtained by solving

$$\bar{X} = \alpha\beta \qquad \frac{1}{n}\sum X_i^2 = \alpha(\alpha + 1)\beta^2$$



$$\hat{\alpha} = \frac{\bar{X}^2}{(1/n)\sum X_i^2 - \bar{X}^2} \qquad \hat{\beta} = \frac{(1/n)\sum X_i^2 - \bar{X}^2}{\bar{X}}$$

152	115	109	94	88	137	152	77	160	165
125	40	128	123	136	101	62	153	83	69

with $\bar{x} = 113.5$ and $(1/20)\sum x_i^2 = 14{,}087.8$. The estimates are

$$\hat{\alpha} = \frac{(113.5)^2}{14{,}087.8 - (113.5)^2} = 10.7 \qquad \hat{\beta} = \frac{14{,}087.8 - (113.5)^2}{113.5} = 10.6$$

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Methods of Point Estimation

- Maximum likelihood estimation（最大似然估计）
- If we have random sample $X_1 \sim X_n$ from a probability distribution, then we naturally want to know the maximum probability for those observed sample data to have occured, and this thought generates the maximum likelihood estimation

Let $X_1, X_2, \ldots, X_n$ have joint pmf or pdf

$$f(x_1, x_2, \ldots, x_n; \theta_1, \ldots, \theta_m) \tag{6.6}$$

where the parameters $\theta_1, \ldots, \theta_m$ have unknown values. When $x_1, \ldots, x_n$ are the observed sample values and (6.6) is regarded as a function of $\theta_1, \ldots, \theta_m$, it is called the **likelihood function.** The maximum likelihood estimates (mle's) $\hat{\theta}_1, \ldots, \hat{\theta}_m$ are those values of the θ_is that maximize the likelihood function, so that

$$f(x_1, \ldots, x_n; \hat{\theta}_1, \ldots, \hat{\theta}_m) \geq f(x_1, \ldots, x_n; \theta_1, \ldots, \theta_m) \quad \text{for all } \theta_1, \ldots, \theta_m$$

When the X_is are substituted in place of the x_is, the **maximum likelihood estimators** result.

Methods of Point Estimation

- A worker checks 10 products, and it has been found that the first, second, and fifth products are defective. If the probability of defective product is dnoted by p, the probability for this sample data to occur is

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- Now we want to know the value of p that makes the probability of this sample data to be maximum, that is, we wish to find the value of p that maximizes this probability

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$$log(f(x_1, x_{2\ldots}x_{10}; p)) = 3 * log(p) + 7 * log(1-p)$$

$$\frac{d}{dp} log(f(x_1, x_{2\ldots}x_{10}; p)) = 3/p - 7/(1-p) = 0$$

$$p = 3/10$$

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$$\frac{d}{dp} log(f(x_1, x_{2\ldots}x_{10}; p)) = 3/p - 7/(1-p) = 0$$

$$p = 3/10$$

- So, the estimator of p is $\hat{p} = 0.3$

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Methods of Point Estimation

Suppose $X_1, X_2, \ldots, X_n$ is a random sample from an exponential distribution with parameter λ. Because of independence, the likelihood function is a product of the individual pdf's:

$$f(x_1, \ldots, x_n; \lambda) = (\lambda e^{-\lambda x_1}) \cdot \cdots \cdot (\lambda e^{-\lambda x_n}) = \lambda^n e^{-\lambda \sum x_i}$$

The ln(likelihood) is

$$\ln[f(x_1, \ldots, x_n; \lambda)] = n \ln(\lambda) - \lambda \sum x_i$$

Equating $(d/d\lambda)$[ln(likelihood)] to zero results in $n/\lambda - \sum x_i = 0$, or $\lambda = n/\sum x_i = 1/\bar{x}$. Thus the mle is $\hat{\lambda} = 1/\bar{X}$;

Methods of Point Estimation

Let $X_1, \ldots, X_n$ be a random sample from a normal distribution. The likelihood function is

$$f(x_1, \ldots, x_n; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_1-\mu)^2/(2\sigma^2)} \cdot \cdots \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_n-\mu)^2/(2\sigma^2)}$$

$$= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} e^{-\sum(x_i-\mu)^2/(2\sigma^2)}$$

so

$$\ln[f(x_1, \ldots, x_n; \mu, \sigma^2)] = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$$

To find the maximizing values of μ and σ^2, we must take the partial derivatives of $\ln(f)$ with respect to μ and σ^2, equate them to zero, and solve the resulting two equations. Omitting the details, the resulting mle's are

$$\hat{\mu} = \overline{X} \qquad \hat{\sigma}^2 = \frac{\sum(X_i - \overline{X})^2}{n}$$

The mle of σ^2 is not the unbiased estimator, so two different principles of estimation (unbiasedness and maximum likelihood) yield two different estimators.

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Methods of Point Estimation

- Summary
- Point estimator and point estimation
  - Point estimate 是具体的估计结果，point estimator 是用来计算这个结果的工具（统计量）
- Unbiased estimator

> A point estimator $\hat{\theta}$ is said to be an **unbiased estimator** of θ if $E(\hat{\theta}) = \theta$ for every possible value of θ. If $\hat{\theta}$ is not unbiased, the difference $E(\hat{\theta}) - \theta$ is called the **bias** of $\hat{\theta}$.

- The important methods:
  - Method of moments
  - Maximum likelihood estimation

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